

Mingyue Xu

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EDUCATION

Purdue University

Ph.D. Student in Computer Science

West Lafayette, IN

Aug 2023 – present

- Advisor: Steve Hanneke

Columbia University

M.S. in Data Science

New York, NY

Sep 2021 – Feb 2023

- Advisor: Daniel Hsu
- GPA: 3.73/4.00

Shanghai Jiao Tong University

B.S. in Mathematics

Shanghai, China

Sep 2017 – Jun 2021

- GPA: 3.62/4.00

RESEARCH INTERESTS

Statistical Learning Theory, Algorithmic Statistics, Deep Learning Theory, Foundations of Artificial Intelligence/LLM

PUBLICATIONS & PRE-PRINTS/UNDER REVIEW

(α - β) DENOTES ALPHABETICAL ORDERING, * DENOTES EQUAL CONTRIBUTION.

- S. Hanneke, H. Wang, **M. Xu** (α - β). “Towards a Theory of Inference-time Alignment with Unknown Rewards”. Preprint.
- D. Hsu and **M. Xu** (α - β). “Attention-based Representations for Multi-task Computation”. Preprint.
- S. Hanneke, H. Wang, **M. Xu** (α - β). “Optimal Learning under Tsybakov Noise”. Preprint.
- S. Hanneke and **M. Xu** (α - β). “When More Data Doesn’t Help: Limits of Adaptation in Multitask Learning”. The Forty-third International Conference on Machine Learning (ICML 2026).
- M. Xu**, G. Vardi, I. Safran. “To Grok Grokking: Provable Grokking in Ridge Regression”. The Forty-third International Conference on Machine Learning (ICML 2026 – **Oral presentation**). **Outstanding Paper Honorable Mention**.
- S. Hanneke and **M. Xu** (α - β). “Universal Rates of ERM for Agnostic Learning”. The Thirty-eighth Annual Conference on Learning Theory (COLT 2025).
- G. Yuan*, **M. Xu***, S. Kpotufe, D. Hsu. “Efficient Estimation of the Central Mean Subspace via Smoothed Gradient Outer Products”. SIAM Journal on Mathematics of Data Science (SIMODS 2025).
- S. Hanneke and **M. Xu** (α - β). “Universal Rates of Empirical Risk Minimization”. The Thirty-eighth Annual Conference on Neural Information Processing Systems (NeurIPS 2024).
- J. Tu, W. Liu, X. Mao, **M. Xu**. “Distributed Semi-supervised Sparse Statistical Inference”. IEEE Transactions on Information Theory (TIT 2023).

WORK EXPERIENCE

Machine Learning Applied Research Scientist Intern
Qutronix Co., Ltd.

Nov 2020 – Jun 2021
Shanghai, China

AWARDS

Outstanding Paper Honorable Mention

Jul 2026

The 43rd International Conference on Machine Learning (ICML 2026)

Excellent Bachelor Thesis Award

Jun 2021

School of Mathematical Sciences, Shanghai Jiao Tong University

Second Prize of Undergraduate Mathematical Contest in Modeling

Oct 2018

Regional Final-Shanghai, China

SERVICE

International Conference on Learning Representations (ICLR) 2026, 2027

Reviewer

International Conference on Machine Learning (ICML) 2026

*Reviewer — **Recognized as a Gold Reviewer***

Neural Information Processing Systems (NeurIPS) 2026

Reviewer

TEACHING EXPERIENCE

CS182 Foundations of Computer Science

Teaching Assistant

Spring 2025, Spring 2026

Purdue University

CS251 Data Structures and Algorithms

Teaching Assistant

Fall 2023, Spring 2024, Fall 2024, Fall 2026

Purdue University

CS381 Introduction to the Analysis of Algorithms

Teaching Assistant

Fall 2025

Purdue University

SKILLS

Python, Java, Pytorch, Deep Learning, Neural Networks, Transformers, Reinforcement Learning, Algorithms and Data Structure, Machine Learning Algorithms, Statistics, Mathematics